Genomic Data Science · DATA405/BIOL480 · Fall 2026 · Howard University

Lab 4 · RNA-seq data analysis

Gene expression with RSEM, Cufflinks and Salmon · Wednesday Oct 28, 2026 · 1.5 hours · Instructor: Jinrui Xu

Name: _____ Lab partner (optional): _____

The question

In 2010, Qian, Liao, Chang and Zhang (Trends in Genetics 26: 425) asked why so many duplicate genes survive in yeast although the two copies do the same job. Their answer: after a gene is duplicated, each copy comes to be expressed at a lower level, so that losing either copy would push the total below what the cell needs. Their evidence was RNA-seq: compared with the single ancestral gene (still single in the fission yeast S. pombe), the two S. cerevisiae copies together are expressed less in 67 % of pairs, versus 52 % for genes that were never duplicated, an excess of about 31 %.

Where did those expression numbers come from? From the Supplementary Methods: the yeast RNA-seq reads of Nagalakshmi et al. (2008), 35-base single-end Illumina reads, counted per gene after mapping with "an algorithm that maps unique (single-copy) sequences to the genome" with up to two mismatches. Reads that fit two places equally well were not counted. For human and mouse they mapped 32- and 25-mers with SeqMap and divided each gene's reads by the number of unique n-mers in the gene (a mappability correction, but still unique reads only).

Here is the catch. Duplicate genes are, by definition, similar to each other. A read from the identical part of two copies fits both, so it is a multi-mapping read, and the unique-only rule throws it away. That happens to duplicates and not to single-copy genes. So a pipeline built on unique reads is biased in exactly the direction of the paper's conclusion. The EM algorithm you met in Lecture 7 exists to solve this: RSEM, Cufflinks and Salmon all share multi-mapping reads between the copies in proportion to the evidence from unique reads. Today you quantify the same reads the paper used, four ways, and decide for yourself whether the conclusion depends on the tool.

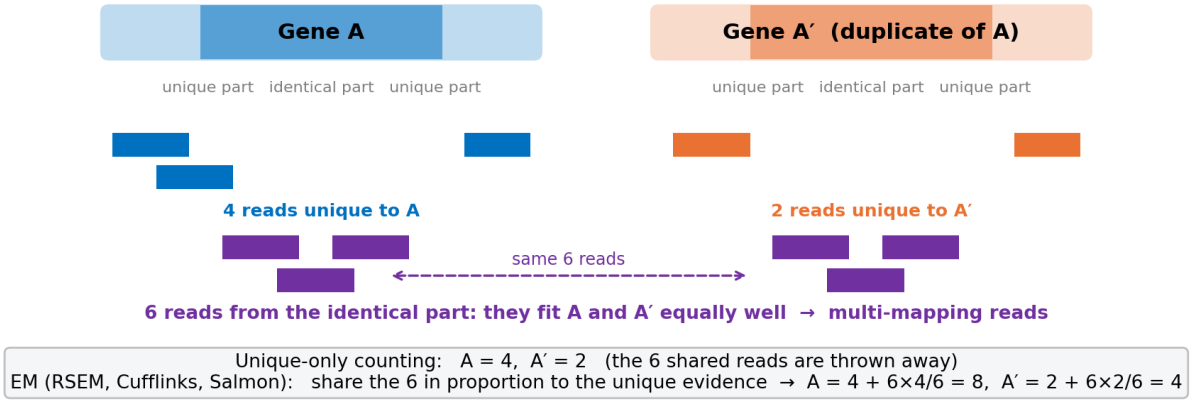


Figure 1. Two copies of a gene share an identical stretch. Reads from it fit both copies.

Key idea Unique-only counting does not make a random error: it removes reads from duplicate genes only, so it makes duplicates look less expressed than they are. Whether that matters depends on how similar the copies are at the DNA level, and on how long the reads are.

What you will do

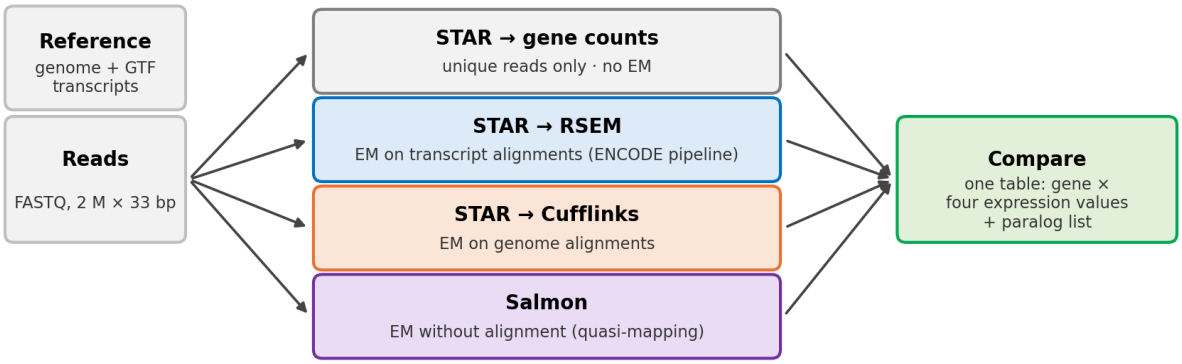


Figure 2. One set of reads, four quantifications. Only STAR's own count uses unique reads alone.

Part	What you do	Where	Time
A	Install the tools; download the reads and reference; run STAR, RSEM, Cufflinks and Salmon	Google Colab notebook (Run all)	≈ 20 min, all waiting: read A1 meanwhile
B	Join the four results with the duplicate-gene table; TEF1/TEF2; the paper's test with and without EM	same notebook (Python), or the course site (Python or R)	≈ 40 min

Before you start

- **The paper:** Qian et al. 2010 and its Supplementary Methods are on the Lab 4 page (PDFs). Read the first two pages and the paragraph of the Supplementary Methods on how the yeast expression levels were obtained.
- **Colab:** open the notebook from the Lab 4 page (https://compbio-geb.github.io/DATA405_2026/lab4.html), File → Save a copy in Drive, then Runtime → Run all. It needs a Google account and nothing else.
- While the tools run (about 20 minutes), read Section A1. Then follow the notebook cell by cell and write down the numbers marked ✏️.
- **Backup:** if Colab fails, the Appendix runs the same four tools by clicking in Galaxy (usegalaxy.eu), and the Lab 4 page has Part B in Python and R using the instructor's results.
- Answer Q1–Q7 and include the two figures. Report due Tuesday, Nov 3, 11:59 pm on Canvas (5 points).

A1 · The ideas you need

Counting reads per gene

RNA-seq measures expression by counting: the more copies of a transcript a cell makes, the more fragments of it are sequenced. After mapping, a gene's expression starts as the number of reads that land in it. Two things must be corrected before genes can be compared (Lecture 7): gene length (a longer transcript collects more reads at the same expression) and sequencing depth (a deeper library gives every gene more reads). TPM (transcripts per million) does both: reads per kilobase of transcript, scaled so that the values of all genes add up to one million. FPKM is the older cousin (fragments per kilobase per million mapped reads). Raw counts are kept too, because statistical tests need them.

Three kinds of read

After mapping, a read is...	Meaning
uniquely mapped	it fits one place in the genome clearly better than any other; counted by every method
multi-mapped	it fits two or more places equally well: identical stretches of duplicate genes, transposons, rRNA repeats; STAR records all the places but counts it for no gene
unmapped	no good match: sequencing errors, adapter, contamination; the 2008 reads used today have poor 3′ ends, so about 40 % are unmapped

What EM does with multi-mapping reads

Suppose gene A has 4 unique reads, its copy A′ has 2, and 6 reads fit both (Figure 1). EM starts from a guess (say, A and A′ are equal), splits each shared read according to the current guess, re-estimates the expression of A and A′ from the split, and repeats until nothing changes. The unique reads anchor the answer: with 4 : 2 unique evidence, the 6 shared reads end up 4 : 2 as well, so A = 8 and A′ = 4. With no unique reads at all (copies that are 100 % identical over their whole length), EM can only split them equally, which is still better than losing them. The three EM tools differ in what they align to:

Tool	Aligns to	Notes
RSEM (2011)	transcript sequences (here via STAR's transcriptome BAM)	the ENCODE and TCGA standard: STAR → RSEM; reports expected_count, TPM, FPKM
Cufflinks (2010)	the genome, with the gene annotation	splits each multi-mapping read evenly between its positions, then EM across isoforms; also assembles transcripts
Salmon (2017)	transcript k-mers, no full alignment (quasi-mapping)	fast, the modern default with kallisto; reports NumReads and TPM
STAR counts (no EM)	the genome	--quantMode GeneCounts: uniquely mapped reads inside one gene; the same rule as the 2010 paper

Today's data

Reads: Nagalakshmi et al. 2008, SRA run SRR002051, the poly(A) RNA-seq library of S. cerevisiae S288C grown in rich medium, 33-bp single-end Illumina Genome Analyzer reads, 2,000,000 of the 3,854,691 reads chosen at random. These are the very reads behind Figure 1 of the 2010 paper. Reference: the R64 (S288C) genome, gene annotation (GTF, 7,127 genes) and the matching transcript sequences from Ensembl Fungi 63. Duplicate genes: every yeast gene's closest paralog from Ensembl, with the protein identity between them (3,091 genes have a paralog; 874 are at least 50 % identical; 326 at least 90 %).

A2 · The notebook, step by step

Every tool step is one cell of shell commands (the lines are what you would type on a Linux server). Open each cell's output as it finishes.

Step 1 · Install (≈ 2 min)

Cell 1

```
apt-get install -y rna-star rsem salmon cufflinks samtools
```

> **What you should see** Four version lines: STAR 2.7.x, RSEM 1.3.x, Salmon 1.x, Cufflinks 2.2.1.

Step 2 · Download (≈ 1 min)

Cell 2

```
wget https://compbio-geb.github.io/DATA405_2026/data/lab4/yeast_reads_2M.fastq.gz      #
2,000,000 reads, 73 MB
wget https://compbio-geb.github.io/DATA405_2026/data/lab4/yeast_genome_R64.fa          # 16
chromosomes + mitochondrion
wget https://compbio-geb.github.io/DATA405_2026/data/lab4/yeast_genes_R64.gtf          # 7,127
genes
wget https://compbio-geb.github.io/DATA405_2026/data/lab4/yeast_transcripts_R64.fa     # their
transcript sequences
wget https://compbio-geb.github.io/DATA405_2026/data/lab4/yeast_paralogs.tsv           #
closest paralog of each gene
```

> **What you should see** The first read is printed: 33 letters and a line of quality characters. 7,127 genes in the GTF.

Step 3 · STAR: map and count unique reads (≈ 10 min)

Cell 3

```
STAR --runMode genomeGenerate --genomeDir star_index --genomeFastaFiles yeast_genome_R64.fa \
     --sjdbGTFfile yeast_genes_R64.gtf --sjdbOverhang 32 --genomeSAindexNbases 10
STAR --genomeDir star_index --readFilesIn yeast_reads_2M.fastq.gz --readFilesCommand zcat \
     --outSAMtype BAM Unsorted --quantMode TranscriptomeSAM GeneCounts --outFileNamePrefix
star_
```

The first command builds the index once (sjdbOverhang = read length − 1; SAindexNbases 10 because the yeast genome is small). The second maps the reads. --quantMode writes (a) counts of uniquely mapped reads per gene, our no-EM baseline and the same rule Qian et al. used, and (b) the alignments in transcript coordinates, which RSEM needs.

> **What you should see** Log: input reads 2,000,000 · uniquely mapped 51.27% · multiple loci 6.78% · too many loci 1.96% · unmapped (too short) 39.28%. ✏️ Record these for Q1. The counts table starts with N_unmapped, N_multimapping, N_noFeature (mapped outside any gene), N_ambiguous; then one row per gene (column 2 = unstranded count).

Step 4 · RSEM: EM on the transcript alignments (≈ 3 min)

Cell 4

```
rsem-prepare-reference --gtf yeast_genes_R64.gtf yeast_genome_R64.fa rsem_ref/yeast
rsem-calculate-expression --alignments --no-bam-output -p 2
star_Aligned.toTranscriptome.out.bam rsem_ref/yeast rsem
```

> **Why this step** RSEM builds its own copy of every transcript from genome + GTF, so it knows exactly which transcripts a read could come from, then runs EM over all reads at once. STAR → RSEM is the ENCODE and TCGA pipeline. Output rsem.genes.results: gene_id, length, effective_length, expected_count (fractional: shared reads are split), TPM, FPKM.

Step 5 · Cufflinks: EM on the genome alignments (≈ 3 min)

Cell 5

```
samtools sort -o star_sorted.bam star_Aligned.out.bam
cufflinks -p 2 -G yeast_genes_R64.gtf -o cufflinks star_sorted.bam
```

> **Why this step** Cufflinks (2010) was the first widely used quantifier. It needs a position-sorted BAM. By default it splits a multi-mapping read evenly between its positions (a read at 2 places counts ½ for each), then runs EM across the isoforms of each gene. It reports FPKM only, no raw count. Its -u option ("multi-read correct") re-weights by unique-read coverage; for near-identical pairs with almost no unique reads it collapses the estimate (TEF1 falls from about 5,000 to 250 FPKM), which is why the lab leaves it off.

Step 6 · Salmon: EM without alignment (≈ 1 min)

Cell 6

```
salmon index -t yeast_transcripts_R64.fa -i salmon_index -k 19
salmon quant -i salmon_index -l U -r yeast_reads_2M.fastq.gz -g yeast_genes_R64.gtf -o salmon
```

> **Why this step** Salmon skips base-by-base alignment: it finds which transcripts share k-mers with the read (k = 19 here, smaller than the default 31 because the reads are 33 bases), scores them, and runs EM. -l U = unstranded; -g adds transcripts up into genes. Mapping rate about 29 %: these error-prone 2008 reads often lack an exact 19-mer.

> **Key idea** Four answers for every gene. STAR's count ignores multi-mapping reads; RSEM and Salmon share them by EM; Cufflinks splits them evenly. The rest of the lab asks when that difference matters.

PART B · COMPARE THE FOUR ANSWERS

The notebook's Part B cells (also on the Lab 4 page, in Python and R, using the instructor's results) read the four result tables and the paralog table and join them into one table d, one row per gene, with these columns:

Column	Meaning
`STAR`	uniquely mapped reads in the gene (no EM)
`RSEM, RSEM_TPM`	RSEM expected count and TPM
`Cufflinks_FPKM`	Cufflinks FPKM
`Salmon, Salmon_TPM`	Salmon reads and TPM
`name, closest_paralog, min_pct_id`	gene name, its closest paralog, and the protein identity between them (%)
`identity`	that identity in four bins: no close paralog (< 30 %), 30–70, 70–95, 95–100 %

Exercise 1 · How many reads did each tool use?

✏️ STAR unique reads in genes _____ RSEM expected counts _____ Salmon reads _____ multi-mapping reads STAR did not count _____

> **What you should see** STAR 681,891 · RSEM 726,466 · Salmon 584,107 · N_multimapping 135,672. RSEM ends up with more reads than STAR's unique count because it keeps the multi-mapping reads; Salmon has fewer because its mapping rate is lower.

Exercise 2 · One duplicate pair: TEF1 and TEF2

TEF1 and TEF2 both encode translation elongation factor 1α; their proteins are 100 % identical and the DNA differs at only a few positions. Fill in your notebook's values (the checkpoint gives the instructor's).

Gene	STAR unique	RSEM	Cufflinks FPKM	Salmon	identity to paralog
TEF1 YPR080W					100 %
TEF2 YBR118W					100 %
TDH3 YGR192C					96.4 %
TDH2 YJR009C					96.4 %
ADH1 YOL086C					93.1 %
ADH2 YMR303C					93.1 %

> **What you should see** TEF1: STAR 768 · RSEM 7,006 · Cufflinks 4,957 FPKM · Salmon 5,585. TEF2: 859 · 6,218 · 4,931 · 4,614. TDH3: 9,111 · 15,613 · 13,750 · 11,719. ADH1: 8,178 · 8,318 · 8,179 · 6,129. Small differences in RSEM, Salmon and Cufflinks are normal (EM starts from a random guess; tool versions).

Exercise 3 · From one pair to all genes

gain = RSEM ÷ STAR for every gene with at least 20 RSEM reads, summarised by identity bin. Fill in the table and keep the box plot for your report.

Identity bin	genes	median gain	75th percentile	share with gain > 1.5
no close paralog				
30–70 %				
70–95 %				
95–100 %				

What you should see 95–100 %: 143 genes, median 1.28, 75th percentile 3.4, 43 % of them gain more than 1.5×. Every other bin: median ≈ 0.9 and at most 11 % gain more than 1.5×. (Median below 1 because STAR's genome count includes reads RSEM does not see, such as reads partly outside the annotated transcript.)

Exercise 4 · The paper's test, with and without EM

For each duplicate pair with at least 40 RSEM reads in total: lost = 1 − ($STAR_1$ + $STAR_2$) ÷ ($RSEM_1$ + $RSEM_2$), the share of the pair's expression that disappears under unique-only counting. The paper's excess of reduced pairs was 31 %.

Identity bin	pairs	median lost	share of pairs losing > 31 %
< 70 %			
70–95 %			
95–100 %			

What you should see 95–100 %: 70 pairs, median lost 0.19, 43 % of pairs lose more than 31 %. 70–95 %: 12 % of 90 pairs. Below 70 %: 2 % of 1,124 pairs. The biggest losers are pairs with no unique read at all, where EM can only split the reads equally: CUP1-1/CUP1-2, ASP3-1…ASP3-4 and HMLα2/MATα2 (100 % identical, 100 % lost), then EFT2/EFT1 (89 %) and TEF2/TEF1 (88 %).

Exercise 5 · Do the three EM tools agree?

What you should see Correlation of log counts: RSEM vs Salmon r ≈ 0.93; RSEM vs STAR unique r ≈ 0.84, with the 95–100 % genes sitting above the diagonal. Cufflinks FPKM vs RSEM TPM r ≈ 0.98.

Report questions (5 points)

Each question 0.5 point; the box plot (Ex 3) and one scatter plot (Ex 5) 0.75 point each.

Q1. From your STAR log: what fraction of the 2,000,000 reads mapped uniquely, mapped to multiple places, and did not map? Give two reasons a 2008 read might not map that would be rarer today.

Q2. Fill in the Exercise 2 table from your notebook for TEF1, TEF2 and TDH3. Which tools agree with each other for TEF1, and which one stands apart? Cufflinks reports FPKM, not a count: explain in one sentence why about 5,000 FPKM and an RSEM count of about 7,000 reads are consistent with each other.

Q3. TEF1 gains about 9× under EM; ADH1 (93 % identical to ADH2) gains almost nothing. Explain the difference in terms of reads of 33 bases: what must be true of a stretch of sequence for a read from it to map uniquely?

Q4. TEF1 and TEF2 ended up with about 7,000 and 6,200 reads under RSEM. Describe in three or four sentences how EM arrived at that split (Lecture 7: what are the hidden variables, the E step and the M step). What would EM do if the two genes were identical over their entire length?

Q5. State in one sentence how Qian et al. counted reads (from the Supplementary Methods summary in Section 1). For which of their duplicate pairs could that rule produce the appearance of reduced expression, and for which could it not? Use your Exercise 4 numbers.

Q6. Does the paper's conclusion survive? Give a verdict with a number: what share of duplicate pairs in each identity bin lose more than the paper's 31 % under unique-only counting, and what does that imply for the whole-genome-duplication pairs (mostly 30–80 % identical) that make up most of their data?

Q7. Modern RNA-seq uses 100–150-base paired-end reads. Predict how the Exercise 4 numbers would change, and whether any pair would still be a problem. Which of the four tools would you choose today, and why?

Figures to include

- Figure A: the box plot of gain by identity bin (Exercise 3).
- Figure B: the RSEM-vs-STAR scatter with the 95–100 % genes highlighted (Exercise 5).

Submit one PDF or Word file on Canvas by Tuesday, Nov 3, 11:59 pm.

The same four steps by clicking in Galaxy

Use usegalaxy.eu (it has RSEM; usegalaxy.org has the other three tools). Sign in, make a new history called Lab 4, then follow the steps. Checked on Oct 27: the numbers are identical to the notebook's.

Step 1 · Upload the four files (≈ 3 min)

1. Click **Upload** → **Paste/Fetch data**. Paste these four addresses, one per line:

Four addresses to paste

```
https://compbio-geb.github.io/DATA405_2026/data/lab4/yeast_reads_2M.fastq.gz
https://compbio-geb.github.io/DATA405_2026/data/lab4/yeast_genome_R64.fa
https://compbio-geb.github.io/DATA405_2026/data/lab4/yeast_genes_R64.gtf
https://compbio-geb.github.io/DATA405_2026/data/lab4/yeast_transcripts_R64.fa
```

2. Galaxy makes one row per address. Set **Type** for each: reads → `fastqsanger.gz`, genome and transcripts → `fasta`, genes → `gtf`. Click **Start**, then **Close**.

> **What you should see** Four green datasets, numbered 1–4 in that order. The reads file is about 73 MB. If a dataset turns red, click its title and the bug icon: the usual cause is a mistyped address.

> **Why this step** A quantification needs three things: the reads, a reference to map them to, and an annotation saying where the genes are. Salmon uses the transcript sequences instead of the genome.

Step 2 · Map with STAR and count unique reads (≈ 5 min)

1. Search **RNA STAR**. Single-end reads: dataset 1.
2. Custom or built-in reference genome: **Use reference genome from history and create temporary index**. Select a reference genome: dataset 2. **Length of the SA pre-indexing string: 10** (the yeast genome is small; the default 14 wastes memory).
3. Build index with or without known splice junctions annotation: **build index with gene-model**. Gene model: dataset 3. Length of the genomic sequence around annotated junctions: **32** (read length − 1).
4. Per gene/transcript output: **Per gene read counts and transcriptome BAM (TranscriptomeSAM GeneCounts)**. Run Tool.

> **What you should see** Five outputs: log, splice junctions, mapped.bam, transcriptome-mapped.bam, reads per gene. Open the log (eye icon): input reads 2,000,000 · uniquely mapped 51.27% · mapped to multiple loci 6.78% · mapped to too many loci 1.96% · unmapped (too short) 39.28%. ✏️ Record these for Q1.

> **Why this step** STAR is the aligner in the ENCODE RNA-seq pipeline. With --quantMode it also writes (a) a count of uniquely mapped reads per gene, which is our no-EM baseline, and (b) the same alignments translated into transcript coordinates, which is what RSEM needs.

Open reads per gene. The first four rows are not genes: N_unmapped, N_multimapping, N_noFeature (mapped outside any gene) and N_ambiguous (inside two overlapping genes). Then one row per gene with three columns: unstranded, forward, reverse. This library is unstranded, so use column 1.

Step 3 · RSEM (≈ 4 min)

1. Search **RSEM calculate expression** (version 1.3.3+galaxy0). Transcriptome alignments: STAR's **transcriptome-mapped.bam**.
2. Reference sequences: **genome** · Genome source: **history** → dataset 2 · Gene annotation: dataset 3 · Library layout: **single**. Run Tool.

> **What you should see** Two tables, gene expression and isoform expression. The gene table has columns gene_id, transcript_id(s), length, effective_length, expected_count, TPM, FPKM. expected_count is the EM estimate of how many reads came from the gene: a decimal, because shared reads are split.

Why this step RSEM builds its own copy of every transcript from the genome + GTF, so it sees exactly which transcripts a read could have come from, then runs EM over all reads at once.

Step 4 · Cufflinks (≈ 4 min)

1. Search **Cufflinks**. SAM or BAM file of aligned RNA-Seq reads: STAR's **mapped.bam**.
2. Use Reference Annotation: **Use reference annotation** → dataset 3. Leave Use multi-read correct at **No**. Run Tool.

What you should see Three outputs; open gene expression: tracking_id (the gene), gene_short_name, length, FPKM and a confidence interval. Cufflinks gives no raw count, only FPKM.

Why this step Cufflinks was the first widely used quantifier (2010). By default it splits a multi-mapping read evenly between its positions (a read at 2 places counts ½ for each), then runs EM across isoforms. Its optional "multi-read correct" re-weights by unique-read coverage; for near-identical pairs that have almost no unique reads it collapses the estimate (TEF1 drops from about 5,000 to 250 FPKM), so leave it off.

Step 5 · Salmon (≈ 3 min)

1. Search **Salmon quant**. Select a reference transcriptome: **Use one from the history** → dataset 4 (the transcripts). Kmer size: **19** (shorter than the default 31 because the reads are only 33 bases).
2. Single-end reads: dataset 1 · library type: **U** (unstranded) · File containing a mapping of transcripts to genes: dataset 3. Run Tool.

What you should see Transcript and gene quantification tables: Name, Length, EffectiveLength, TPM, NumReads. The mapping rate in the log is about 29 %: Salmon needs an exact 19-mer seed, which many of these error-prone old reads lack.

Why this step Salmon skips base-by-base alignment: it finds which transcripts share k-mers with the read, scores them, and runs EM. On human data it takes minutes where STAR + RSEM take an hour.

Glossary

Term	Meaning
Read count	number of reads assigned to a gene; the raw measure of expression
TPM	transcripts per million: reads per kilobase, scaled so all genes sum to 10^6; comparable across genes and samples
FPKM / RPKM	fragments (reads) per kilobase per million mapped reads; older length-and-depth normalisation
Multi-mapping read	a read that aligns equally well to two or more places
EM algorithm	expectation–maximisation: alternate between assigning shared reads given current expression (E) and re-estimating expression from the assignment (M) until convergence
expected_count	RSEM's EM estimate of reads per gene; fractional because shared reads are split
Paralogs	genes in one genome descended from a duplication; ohnologs are paralogs from the whole-genome duplication ~100 Myr ago in yeast
Protein identity	% of aligned amino acids that are the same between two paralogs; the DNA identity is usually a little lower
Quasi-mapping	Salmon's way of placing reads on transcripts by shared k-mers without full alignment
BAM	binary file of read alignments; a transcriptome BAM has the same reads in transcript coordinates

Sources: Qian W, Liao B-Y, Chang AY-F, Zhang J (2010) Trends Genet 26: 425–430, and its Supplementary Methods · Nagalakshmi U et al. (2008) Science 320: 1344 (reads: GEO GSE11209, SRA SRR002051) · Li B & Dewey C (2011) RSEM, BMC Bioinformatics 12: 323 · Trapnell C et al. (2010) Cufflinks, Nat Biotechnol 28: 511 · Patro R et al. (2017) Salmon, Nat Methods 14: 417 · Dobin A et al. (2013) STAR, Bioinformatics 29: 15 · Ensembl Fungi release 63 (R64-1-1).